Where to Spend the Next Ad Dollar

Facebook Ads vs. Google AdWords

Marketing Performance Analysis & Budget Allocation Strategy

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PROJECT TYPE	DATASET COVERAGE	ANALYTICAL FOCUS
Self-directed portfolio case study	1,000 records · 22 Dec 2021–20 Dec 2024 · 658 distinct dates	Channel performance, statistical comparison and budget allocation

Executive summary

Facebook recorded a lower aggregate cost per recorded conversion than Google AdWords in the available dataset. Google AdWords generated greater reach and click volume at lower CPC and CPM. The findings indicate different strengths across the marketing funnel and support an objective-led channel strategy.

This is an observational analysis of the supplied records. The dataset does not document randomized assignment, comparable audience groups, attribution rules, currency or revenue. The results therefore do not establish that either platform caused better outcomes or generated greater financial return.

Facebook Ads	Google AdWords
Stronger on recorded conversions, conversion rate and cost per recorded conversion.	Stronger on views, clicks, cost per click and cost per 1,000 views.
Aggregate CPA: 13.08 dataset units.	Aggregate CPA: 21.02 dataset units.

1. Key performance snapshot

The dataset contains 1,000 records from 22 December 2021 to 20 December 2024, covering 658 distinct dates.

Metric	Facebook Ads	Google AdWords	Advantage
Views	2,152,031	4,771,438	Google
Clicks	44,196	60,128	Google
Recorded conversions	11,975	5,933	Facebook
Recorded advertising cost*	156,610	124,741	Google (lower total)
Click-through rate (CTR)	2.05%	1.26%	Facebook
Conversion rate (CVR)	27.10%	9.87%	Facebook
Cost per click (CPC)*	3.54	2.07	Google
Cost per 1,000 views (CPM)*	72.77	26.14	Google
Cost per recorded conversion (CPA)*	13.08	21.02	Facebook

** Cost values are in unspecified dataset units; the source does not identify a currency. CPA means cost per recorded conversion and should not automatically be interpreted as cost per unique customer or sale.*

2. Interpretation of findings

Facebook: stronger recorded conversion efficiency

Facebook recorded 11,975 conversions compared with 5,933 for AdWords—approximately 102% more. Its aggregate CPA was 13.08 versus 21.02 dataset units, approximately 38% lower. Facebook's aggregate CVR was also higher (27.10% versus 9.87%). These are descriptive comparisons of the supplied records.

Google AdWords: stronger reach and traffic efficiency

AdWords recorded approximately 2.2 times as many views and more clicks. Its CPC was approximately 41% lower (2.07 versus 3.54), and its CPM was approximately 64% lower (26.14 versus 72.77). These results may make AdWords useful for reach and traffic objectives.

Traffic volume is not the same as conversion outcome

AdWords generated 60,128 clicks compared with Facebook's 44,196, but recorded fewer than half as many conversions. Channel performance should therefore be evaluated against campaign objectives and, where available, downstream conversion quality.

3. Statistical analysis

The date-level paired comparison reports a mean CPA difference (Facebook minus AdWords) of −8.50 dataset units, with a 95% confidence interval of −9.52 to −7.48. The paired t-test and Wilcoxon signed-rank test both report $p < 0.001$.

These results indicate a statistically detectable difference in the observed date-level CPA values under the aggregation and pairing assumptions. They do not establish causation. The dataset does not fully document the observation unit, repeated-date structure or attribution rules, so the statistical findings should be interpreted with caution.

4. Recommended budget pilot

For a conversion-focused objective, a controlled 4–6 week pilot could evaluate reallocating 15% of the eligible AdWords conversion budget to Facebook. This is a proposed business action to test—not a proven outcome of the historical analysis.

Pilot element	Proposed approach
Budget under evaluation	Approximately 18,711 dataset units, representing 15% of the recorded AdWords cost benchmark (124,741); confirm the actual conversion-focused budget before implementation.
Duration	4–6 weeks, using comparable measurement periods.
Primary measure	Cost per recorded conversion on incremental Facebook spend.
Proposed pass threshold	CPA at or below 16 dataset units; a proposed decision threshold, not a statistically derived benchmark.
Stop threshold	CPA above 21.02 dataset units, the historical AdWords aggregate CPA.
Channel role	Retain AdWords for reach and traffic objectives while evaluating Facebook for conversion-focused activity.

Pilot design: Use comparable audience definitions, conversion events, attribution windows, budgets and measurement periods. Where feasible, use randomized audience groups or a suitable holdout. Define the primary KPI, minimum conversion volume and stop rules before the pilot begins.

5. Illustrative scenario—not a forecast

If approximately 18,711 dataset units generated conversions at Facebook's historical CPA of 13.08, the arithmetic implies about 1,430 recorded conversions. At AdWords' historical CPA of 21.02, the same amount implies about 890. The difference is approximately 540 conversions.

This is an illustrative best-case calculation using historical average CPAs, not a forecast or guaranteed incremental gain. Actual results may differ as budgets, audiences and marginal returns change.

6. Confidence and decision guardrails

Finding	Confidence	Basis
Facebook has lower aggregate CPA	Moderate	Observed in the dataset, but cost definition, attribution and record structure are unclear.
AdWords has lower CPC and CPM	High for descriptive totals	Based on aggregate counts and recorded cost; cost units remain unspecified.
Budget reallocation will increase conversions	Low	Incremental returns and audience overlap are unknown; a controlled pilot is needed.
Facebook causes better results	Not established	No random assignment or valid experimental design is documented.

7. Limitations and next steps

- Observational data: no documented random assignment, audience split or controlled experimental design.
- Unclear observation grain: 1,000 records cover 658 dates, with multiple records on many dates; the reason and aggregation rules are not documented.
- Attribution and conversion quality: ten Facebook records contain more recorded conversions than clicks; these are retained and flagged, but the attribution explanation is unknown.
- Cost definition: no currency or detailed cost-field definition is supplied.
- No revenue or margin: ROAS, profitability and customer lifetime value cannot be assessed.
- The budget pilot should be evaluated using incremental results and comparable measurement conditions, not historical averages alone.

Conclusion

Within the available records, Facebook is stronger on recorded conversion volume and aggregate CPA, while Google AdWords is stronger on reach and traffic efficiency. The business recommendation is to align channel investment with campaign objectives, validate measurement definitions and use a controlled pilot to determine whether additional Facebook investment delivers incremental value. The analysis does not establish causal superiority or profitability.

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